

Ishaan Agarwal

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Education

Cornell University, Ann S. Bowers College of Computing and Information Science

Ithaca, NY

Master of Professional Studies, Applied Statistics – GPA: 3.87/4.0

May 2026

- Coursework: Data Mining & Machine Learning, Applied Neural Networks, Introduction to Deep Learning, Bayesian Data Analysis, Applied Time Series Analysis, Modern Regression, Categorical Data, Statistical Computing.

Manipal Institute of Technology

Karnataka, India

B.Tech, Electronics and Communication Engineering – Minor: Data Science

June 2023

Experience

UBS Bank India

Software Engineer, Data & Search Infrastructure (Internship → Full-Time)

January 2023 – July 2025

- Built and maintained Java/Spring Boot microservices powering 20+ Azure Search indexes for UBS Neo's cross-asset trading platform, serving 1.8M+ users and 200K daily API calls with real-time ingestion from Kafka, Tibco, and REST APIs.
- Prototyped a RAG pipeline auto-tagging 10,000+ financial documents via LLM-generated tags from semantically similar context, cutting manual tagging effort 60%; placed Top 5 in UBS AI Quest.
- Led migration of a business-critical search service to cloud-native AKS, replacing service-level auth with platform-wide verification; shipped with zero production incidents.
- Extended the Azure Search schema for region-based FX data segmentation and executed a full reindex of 2.4M+ financial instrument records.
- Migrated core services from 29West to gRPC and Azure Cache for Redis, enabling AKS deployment and a real-time data-quality monitoring pipeline across upstream sources.

Research

Embedding-Based Causal Adjustment for Latent Confounders

Cornell University

Advisor: Prof. Jelena Bradic – Cornell SDS Project Showcase

January – May 2026

- Investigated whether LLM-derived embeddings of household histories serve as valid proxies for latent confounders when estimating the causal effect of job training on earnings.
- Designed a semi-synthetic benchmark on 6,107 PSID households with confounders across varied outcome and treatment regimes.
- Built and benchmarked 7 embedding pipelines spanning TF-IDF+SVD, Sentence-Transformers, trajectory, fusion, adversarial, and contrastive approaches.
- Estimated Average Treatment Effects via Double Machine Learning with AIPW and cross-fitting, reducing RMSE by 2–3.4× and bias by 55% versus covariate-only adjustment; applied to real PSID data (waves 2007–2021), finding no statistically significant effect of job training on income.

Projects

From-Scratch RAG System with Eval & Observability Console

FastAPI, ChromaDB, Ollama, React, TypeScript

2026

- Built a full-stack RAG system from scratch (FastAPI, ChromaDB, local LLMs via Ollama) with cited answers, a config-versioned pipeline, and a React/TypeScript observability console; 135 automated tests across backend and frontend, developed via TDD.
- Designed an LLM-as-judge evaluation framework scoring 3 quality metrics per query (context relevance, faithfulness, answer relevance) with automated failure attribution across retrieval vs. generation, reaching 83% agreement with hand-labeled examples.
- Instrumented per-query tracing (retrieved chunks with similarity scores, exact prompts, per-stage latency, token counts) tagged with a config-version hash, making answer-quality regressions attributable to specific pipeline changes.

GitHub Archive Dual-Path ETL Pipeline

PySpark, Airflow, Kafka, PostgreSQL

- Built a dual-path (streaming + batch) ETL pipeline over GitHub Archive event data using PySpark and Airflow, ingesting 267,000+ events/hour and curating 112,000+ rows per run into PostgreSQL.
- Designed YAML-driven dynamic DAG generation in Airflow to parameterize pipeline configs without code changes, with built-in data-quality gates validating schema and completeness before load.

Skills

Languages & Libraries:	Python, Java, Spring Boot, R, SQL, PyTorch, Scikit-learn, HuggingFace, Pandas, NumPy
ML & Statistics:	Causal Inference (DML, AIPW), Time Series Forecasting, Deep Learning, RAG, LLM Embeddings, Computer Vision, High-Dimensional Statistics
Data & Infra:	PySpark, Airflow, Kafka, Azure (AKS, Search, Redis), Kubernetes, GitLab CI/CD, PostgreSQL, ChromaDB, FastAPI, REST APIs
Certifications:	Microsoft Azure AI Fundamentals (AI-900); Deep Learning Specialization (DeepLearning.AI)